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MIL-G-29170B
28 OCTOBER 1988
SUPERSEDING
MIL-G-29170A(YD)
17 December 1982

MILITARY SPECIFICATION

GASKETS, SYNTHETIC RUBBER, REINFORCED (FOR DRYDOCK CAISSONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers molded, reinforced, medium-soft, oil-resistant, synthetic rubber gaskets for drydock caissons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat Panel.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A

|FSC 5330|

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-190 - Identification Marking of Rubber Products.

(Unless otherwise indicated, copies of Federal and Military specifications and standards are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 395 - Standard Test Methods for Rubber Properties - Compression Set (DoD adopted).
ASTM D 412 - Standard Test Methods for Rubber Properties in Tension (DoD adopted).
ASTM D 413 - Standard Test Method for Rubber Properties - Adhesion to Flexible Substrate (DoD adopted).
ASTM D 471 - Standard Test Method for Rubber Properties - Effect of Liquids (DoD adopted).
ASTM D 572 - Standard Test Method for Rubber Properties - Deterioration by Heat and Oxygen (DoD adopted).
ASTM D 2240 - Standard Test Method for Rubber Properties - Durometer Hardness (DoD adopted).
ASTM D 3951 - Standard Practice for Commercial Packaging (DoD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The gasket shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3 Construction.

3.3.1 Reinforcing material. Unless otherwise specified (see 6.2), reinforcing material shall be six plies of synthetic woven fabric or cotton duck as shown in figure 1. The reinforcing material shall be firmly and evenly woven, and free from unsightly defects, knots, lumps, dirt, or irregularities of twist. The fabric shall be of a basket or plain weave and shall weigh not less than 18.0 ounces per square yard. The warp and filler shall each have not less than 21 threads per linear inch. The fabric shall have a tensile strength of not less than 260 pounds both in the direction of the warp and the fill.

3.3.2 Rubber compound. The gasket shall be molded of synthetic rubber compound and when vulcanized shall conform to the requirements specified in table I.

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TABLE I. Physical requirements.

Characteristics of rubber compound	Requirements
Tensile strength	Not less than 2,500 pounds per square inch (psi)
Tensile stress at 300 percent elongation	Not less than 1,200 psi
Tensile strength after aging in oxygen bomb for 48 hours	Not less than 70 percent of tensile strength before aging
Ultimate elongation	Not less than 400 percent
Ultimate elongation after aging in in oxygen bomb for 48 hours	Not less than 70 percent of elongation before aging
Compression set	Not greater than 15 percent
Adhesion strength between rubber and between fabric plies	Not less than 15 pounds per inch
Hardness, Shore A, 3-second:	
Minimum	Not less than 55
Maximum	Not greater than 65
Properties after immersion in test liquid:	
Tensile strength, maximum change	Not greater than 40 percent
Ultimate elongation, maximum change	Not greater than 33 percent
Tensile stress at 300 percent elongation, maximum change	Not greater than 30 percent
Volume, maximum change	Not greater than 30 percent

3.3.3 Dimensions. The length, width, height, and thickness of the gasket shall be as specified (see 6.2). Dimensional tolerances shall be as specified in table II.

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TABLE II. Dimensional tolerances.

Dimension	Tolerances in cross-sectional dimension
Flange thickness:	
When 0.625 inch or less	±0.03125 inch
When greater than 0.625 inch	±0.0625 inch
Overall height: 1/	
When 2.0 inches or less	±0.0625 inch
When greater than 2.0 inches	±0.09375 inch
Overall width:	
When 6.0 inches or less	±0.0625 inch
When greater than 6.0 inches	±0.125 inch

1/ Overall height shall not differ greater than 0.03125 inch when measured at any two points which are 2.0 inches apart along the length of the gasket.

3.4 Identification marking. The gasket shall be marked in accordance with MIL-STD-190. Special markings, when required, shall be as specified (see 6.2).

3.5 Workmanship. Workmanship shall be such that the product shall be free from imperfections which will adversely affect its general appearance, function, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of

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manufacturing operations, is an acceptable practic to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All gaskets of the same size and length offered for delivery at one time shall be considered a lot for the purpose of inspection (see 4.3.2). If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.0 percent defective for major defects and an AQL of 2.5 defective for minor defects.

4.3.2 Sampling for tests. In addition to the samples selected in accordance with 4.3, each lot of gaskets shall include the following materials for test purposes:

a. Not less than a 24.0-inch section of rubber gasket identical to the gasket submitted for the tests.
b. Not less than 12.0 inches of unfrictioned synthetic woven fabric or cotton duck of the full width of the bolt or roll used in fabricating the gasket lot.

4.4 Examination. Each sample selected in accordance with 4.3 shall be examined for defects listed in table III. Each attribute within each classification of multiple defects shall constitute one defect.

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TABLE III. Classification of defects.

Classification	Defects	Requirement paragraph
Major:		
101	Dimensions and dimensional tolerances not as specified	3.3.3 and table II
102	Surfaces not smooth	3.3.1 and 3.5
Minor:		
201	Marking not as specified	3.4

4.5 Tests. Each sample and additional material selected in accordance with 4.3 and 4.3.2, shall be subjected to the tests specified in table IV to determine conformance to 3.3.1 and 3.3.2. Failure of any sample to pass the tests of table IV, as applicable, shall be cause for rejection and the lot represented by the sample shall be rejected. Unless otherwise specified (see 6.2), the test conditions shall be as specified in the applicable test method.

TABLE IV. Test methods.

Test	Test method
Tensile strength	ASTM D 412, Method A using straight specimens
Tensile stress at 300 percent elongation	ASTM D 412, Method A using straight specimens
Ultimate elongation	ASTM D 412, Method A using straight specimens
Accelerated aging	ASTM D 572
Compression set	ASTM D 395, Method C
Adhesion	ASTM D 413, Machine method
Hardness	ASTM D 2240, using Type A durometer

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TABLE IV. Test methods - Continued.

Test	Test method
Immersion:	
Tensile strength after immersion	ASTM D 471, test fluid shall be water
Ultimate elongation	ASTM D 471, test fluid shall be water
Tensile stress at 300 percent	ASTM D 471, test fluid shall be water
Volume change	ASTM D 471, test fluid shall be water
Thread count	FED-STD-191, Method 5050
Tensile strength of fabric reinforcement	FED-STD-191, Method 5100

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Gaskets shall be preserved method IA in accordance with MIL-P-116. Use stiffening material internally if needed to maintain rigidity.

5.1.2 Commercial. Gaskets shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Gaskets shall be packed in snug-filling wood-cleated fiberboard, cleated plywood, nailed wood, or exterior grade fiberboard box, conforming to PPP-B-591, PPP-B-601 (overseas), PPP-B-621, or PPP-B-640. Plywood shall conform to type 1 or 2, class 2 of NN-P-530. Containers shall be lined with sealed waterproof case liner conforming to MIL-L-10547 and appendix thereto. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood or wood-cleated boxes shall not be greater than 70 pounds.

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5.2.2 Level B. Gaskets shall be packed as specified in 5.2.1, except domestic type wood boxes or boxes conforming to PPP-B-636 shall be used.

5.2.3 Commercial. Gaskets shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any special marking specified in the contract (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gaskets covered by this specification are intended for use on the seating surfaces of drydock caissons.

6.2 Acquisition regirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
c. Reinforcing material, if other than as specified (see 3.3.1).
d. Length, width, height, and thickness of gasket (see 3.3.3).
e. Special markings for gasket, if required (see 3.4).
f. Test conditions for testing the gasket and material, if other than as specified (see 4.5).
g. Level of preservation, and level of packing (see 5.1 and 5.2).
h. Special markings for shipping containers, if required (see 5.3).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements list (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 Subject term (key word) listing.

Cotton duck
Drydock caissons
Gaskets, synthetic rubber, reinforced
Synthetic woven fabric

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activity:
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Preparing activity:
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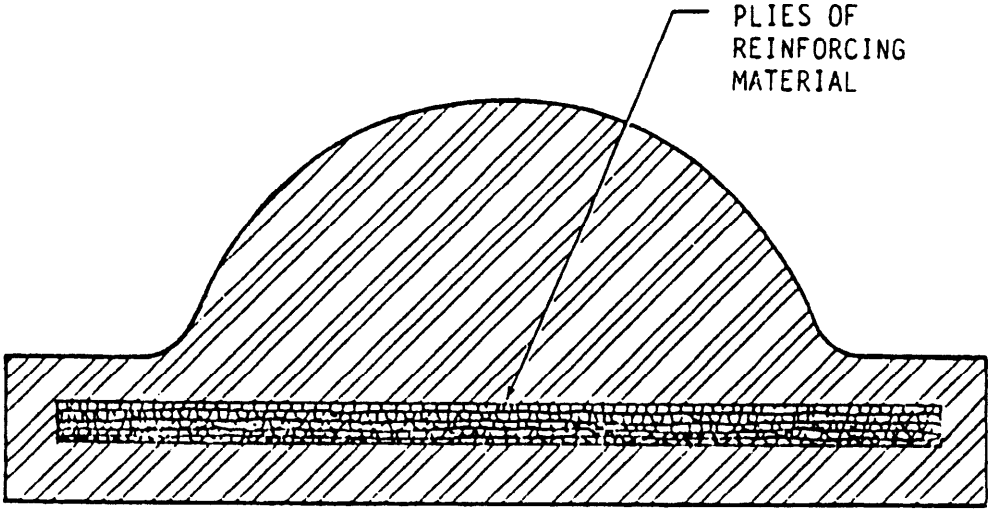


FIGURE 1. Cross-section of rubber gasket.

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STANDARDIZATION DOCU... IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-29170B	GASKETS, SYNTHETIC RUBBER, REINFORCED (FOR DRYLOCK CAISSONS)

3a. NAME OF SUBMITTING ORGANIZATION

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

b. ADDRESS *(Street, City, State, ZIP Code)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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